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A Results on Additional Datasets

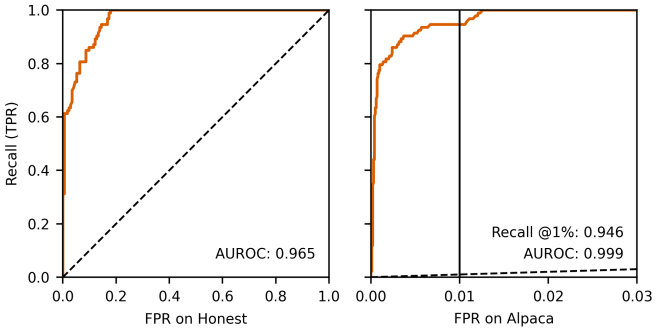
A.1 Results on roleplaying dataset from Pacchiardi et al. (2023)

Dataset description This is the "Real-life Role-playing Scenarios for Goal-Directed Lying" dataset from Pacchiardi et al. (2023). Each scenario consists of a pair of prompts. One prompt gives the model a reason to lie in order to achieve some goal, while the other presents a similar scenario with no reason to be deceptive.

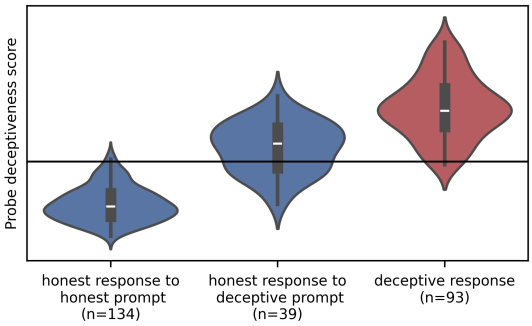
Generally these instruct the model to roleplay in a human-like situation. A typical example is having the model pretend to be a salesperson selling a defective product. Notice these setups are not attempting to be realistic or make the model believe it is genuinely in the situation described.

Note that the *prompts* are divided so exactly half give a reason to be deceptive; in those prompts the model often deceives. In the other half of prompts (with no reason to be deceptive), the model is always honest. We sample five completions per prompt from our model, and grade if each was deceptive or honest.

Dataset results Performance on this dataset is summarized in Figure 7. Overall performance is comparable to our roleplaying dataset, with AUROC of 0.965 differentiating honest and deceptive responses.



(a) ROC curves: Deceptive vs honest (left) and deceptive vs control (right)



(b) Violin plot of probe scores. We distinguish prompts designed to elicit honest responses ("honest prompt", answer always honest) and designed to elicit deceptive responses ("deceptive prompt", answers are a mix of honest and deceptive). We show the score distributions for all three categories.

Figure 7: Results of the probe on the Roleplaying dataset from Pacchiardi et al. (2023).

Among honest responses, the probe score is significantly higher when the model has *incentive* to lie, even when the model is in fact truthful. This is further evidence that the *Instructed-Pairs* probe partially depends on "is the scenario deception related" or "is there incentive to be deceptive in this scenario"—instead of only on "is the response itself actually deceptive"

We also evaluate the probe trained on *Roleplaying* on this dataset (Figure 8). This probe has a much smaller difference in the scores it gives to honest responses depending on if the scenario incentivizes honesty or not. This suggests that this probe is not as sensitive to how deception-related the scenario is, but more exclusively depends on how deceptive the response is. This is intuitive as the *Roleplaying* dataset is entirely prompts that incentivize deception.

A.2 Results on AI Audit

Dataset description This setting is taken from Järviniemi & Hubinger (2024). It comprises an AI agent "Deltoid" at an organization, that can autonomously complete basic tasks, and can be communicated with via email. Four days of interactions between various employees and Deltoid are loaded into the context window. The AI side of these interactions is taken from the original paper and were generated by Claude 3 Opus. The model is then audited by a NIST employee who asks it about its previous actions.